

**SAVEETHA SCHOOL OF ENGINEERING**  
**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES**  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

|                      |                                  |                     |           |
|----------------------|----------------------------------|---------------------|-----------|
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| <b>COURSE NAME:</b>  | STATISTICS WITH R<br>PROGRAMMING | <b>COURSE CODE:</b> | ITA0404   |

**COURSE OUTCOME COVERED: CO3 — ASSESSMENT TOOL 1**

## 1. Medical – Patient Health Analysis

A hospital records the number of patients visiting different departments over three months.

| Department  | January | February | March |
|-------------|---------|----------|-------|
| Cardiology  | 120     | 135      | 150   |
| Neurology   | 90      | 105      | 110   |
| Orthopedics | 100     | 115      | 125   |

### Tasks:

- Find the total patients for each department.
- Identify the department with the highest number of patients.
- Melt the data into long format.
- Cast the long-format data back into wide format.

### R SOLUTION CODE:

```
# Load required library
library(reshape2)

# 1. Create Data Frame
medical_data <- data.frame(
  Department = c("Cardiology", "Neurology", "Orthopedics"),
  January = c(120, 90, 100),
  February = c(135, 105, 115),
  March = c(150, 110, 125)
)

# 2. Total patients for each department
medical_data$Total_Patients <- rowSums(medical_data[, c("January", "February", "March")])
print("Data Frame with Total Patients:")
print(medical_data)

# 3. Department with highest number of patients
max_dept <- medical_data$Department[which.max(medical_data$Total_Patients)]
cat(
  "Department with highest number of patients:", max_dept, "
")

# 4. Melt the data into long format
medical_long <- melt(medical_data[, 1:4], id.vars = "Department",
  variable.name = "Month", value.name = "Patients")
print("Long Format (Melted):")
print(medical_long)

# 5. Cast long-format data back into wide format
medical_wide <- dcast(medical_long, Department ~ Month, value.var = "Patients")
print("Wide Format (Casted):")
print(medical_wide)
```

## 2. Smart City – Electricity Consumption Analysis

A smart city monitors electricity consumption in different zones.

| Zone   | Monday | Tuesday | Wednesday |
|--------|--------|---------|-----------|
| Zone-A | 450    | 470     | 490       |
| Zone-B | 520    | 510     | 550       |
| Zone-C | 380    | 400     | 420       |

### Tasks:

- Calculate total electricity consumption for each zone.
- Identify the zone with the highest consumption.
- Convert the data from wide to long format using melting.
- Convert it back to wide format using casting.

### R SOLUTION CODE:

```
# Load required library
library(reshape2)

# 1. Create Data Frame
elec_data <- data.frame(
  Zone = c("Zone-A", "Zone-B", "Zone-C"),
  Monday = c(450, 520, 380),
  Tuesday = c(470, 510, 400),
  Wednesday = c(490, 550, 420)
)

# 2. Calculate total electricity consumption for each zone
elec_data$Total_Consumption <- rowSums(elec_data[, c("Monday", "Tuesday", "Wednesday")])
print("Data Frame with Total Consumption:")
print(elec_data)

# 3. Zone with highest consumption
max_zone <- elec_data$Zone[which.max(elec_data$Total_Consumption)]
cat(
  "Zone with highest consumption:", max_zone, "
")

# 4. Convert wide to long format (Melting)
elec_long <- melt(elec_data[, 1:4], id.vars = "Zone",
  variable.name = "Day", value.name = "Consumption")
print("Long Format (Melted):")
print(elec_long)

# 5. Convert long back to wide format (Casting)
elec_wide <- dcast(elec_long, Zone ~ Day, value.var = "Consumption")
print("Wide Format (Casted):")
print(elec_wide)
```

### 3. Traffic – Vehicle Flow Analysis

A smart traffic system records the number of vehicles passing through three junctions.

| Junction   | Morning | Afternoon | Evening |
|------------|---------|-----------|---------|
| Junction-1 | 850     | 620       | 950     |
| Junction-2 | 720     | 580       | 810     |
| Junction-3 | 950     | 700       | 1050    |

#### Tasks:

- Create the data frame.
- Find the total vehicles recorded at each junction.
- Identify the time period with the highest traffic.
- Melt the data into long format.
- Cast the data back into wide format.

#### R SOLUTION CODE:

```
# Load required library
library(reshape2)

# 1. Create Data Frame
traffic_data <- data.frame(
  Junction = c("Junction-1", "Junction-2", "Junction-3"),
  Morning = c(850, 720, 950),
  Afternoon = c(620, 580, 700),
  Evening = c(950, 810, 1050)
)

# 2. Total vehicles recorded at each junction
traffic_data$Total_Vehicles <- rowSums(traffic_data[, c("Morning", "Afternoon", "Evening")])
print("Data Frame with Total Vehicles:")
print(traffic_data)

# 3. Identify time period with highest traffic
time_totals <- colSums(traffic_data[, c("Morning", "Afternoon", "Evening")])
highest_period <- names(time_totals)[which.max(time_totals)]
cat(
  "Time period with highest traffic:", highest_period, "
")

# 4. Melt data into long format
traffic_long <- melt(traffic_data[, 1:4], id.vars = "Junction",
  variable.name = "Time_Period", value.name = "Vehicles")
print("Long Format (Melted):")
print(traffic_long)

# 5. Cast data back to wide format
traffic_wide <- dcast(traffic_long, Junction ~ Time_Period, value.var = "Vehicles")
print("Wide Format (Casted):")
print(traffic_wide)
```

## 4. Agriculture – Crop Production Analysis

An agricultural department records crop production in different districts.

| District  | Rice | Wheat | Maize |
|-----------|------|-------|-------|
| Salem     | 450  | 300   | 250   |
| Erode     | 520  | 280   | 310   |
| Thanjavur | 700  | 350   | 290   |

### Tasks:

- Calculate total production for each district.
- Identify the district with the highest production.
- Melt the data to obtain District, Crop, and Production.
- Cast the melted data to its original structure.

### R SOLUTION CODE:

```
# Load required library
library(reshape2)

# 1. Create Data Frame
crop_data <- data.frame(
  District = c("Salem", "Erode", "Thanjavur"),
  Rice = c(450, 520, 700),
  Wheat = c(300, 280, 350),
  Maize = c(250, 310, 290)
)

# 2. Total production for each district
crop_data$Total_Production <- rowSums(crop_data[, c("Rice", "Wheat", "Maize")])
print("Data Frame with Total Production:")
print(crop_data)

# 3. District with highest production
max_district <- crop_data$District[which.max(crop_data$Total_Production)]
cat("
District with highest production:", max_district, "
")

# 4. Melt data to obtain District, Crop, and Production
crop_long <- melt(crop_data[, 1:4], id.vars = "District",
  variable.name = "Crop", value.name = "Production")
print("Long Format (Melted):")
print(crop_long)

# 5. Cast melted data to original structure
crop_wide <- dcast(crop_long, District ~ Crop, value.var = "Production")
print("Wide Format (Casted):")
print(crop_wide)
```

## 5. Smart Agriculture – Soil Monitoring Analysis

An IoT-based agricultural system records soil parameters for different fields.

| Field   | Moisture | Temperature | pH  |
|---------|----------|-------------|-----|
| Field-A | 65       | 28          | 6.5 |
| Field-B | 55       | 30          | 7.0 |
| Field-C | 72       | 27          | 6.2 |

### Tasks:

- Create the data frame.
- Identify the field with the highest soil moisture.
- Calculate the average temperature.
- Reshape the data into long format using melting.
- Cast the data back into wide format.

### R SOLUTION CODE:

```
# Load required library
library(reshape2)

# 1. Create Data Frame
soil_data <- data.frame(
  Field = c("Field-A", "Field-B", "Field-C"),
  Moisture = c(65, 55, 72),
  Temperature = c(28, 30, 27),
  pH = c(6.5, 7.0, 6.2)
)
print("Original Data Frame:")
print(soil_data)

# 2. Field with highest soil moisture
highest_moisture_field <- soil_data$Field[which.max(soil_data$Moisture)]
cat("
Field with highest soil moisture:", highest_moisture_field, "
")

# 3. Calculate average temperature
avg_temp <- mean(soil_data$Temperature)
cat("Average Temperature:", avg_temp, "
")

# 4. Reshape into long format using melting
soil_long <- melt(soil_data, id.vars = "Field",
  variable.name = "Parameter", value.name = "Value")
print("Long Format (Melted):")
print(soil_long)

# 5. Cast data back into wide format
soil_wide <- dcast(soil_long, Field ~ Parameter, value.var = "Value")
print("Wide Format (Casted):")
print(soil_wide)
```